

# Supervised ML (Assignment2)

## Assignment Project

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### Problem Statement

A multinational company **TalentCore Pvt. Ltd.** has been experiencing a rising number of employee resignations, which increases recruitment cost, project delays, and loss of skilled talent.

The HR department wants to build an **intelligent ML system** that can **predict whether an employee is likely to leave the company** based on job satisfaction, salary, age, work-life balance, training hours, bonuses, and other work-related factors.

You are hired as a **AI/ML Engineer** to:

1. Build a **baseline Logistic Regression model**
2. Improve it using **Regularization (L1 & L2)**
3. Compare their performances and recommend the best model.

### Dataset Description

The Employee dataset has 900 rows and 15 columns (features), representing various employee metrics. This dataset aims to reflect realistic scenarios in a corporate settings.

Following is a brief overview of the dataset:

| Feature                                 | Description                                                  |
|-----------------------------------------|--------------------------------------------------------------|
| Job_Satisfaction                        | Level of satisfaction with the job                           |
| Performance_Rating                      | Employee performance score                                   |
| Years_At_Company                        | Number of years worked in company                            |
| Work_Life_Balance                       | Balance between work and personal life                       |
| Distance_From_Home                      | Distance of home from workplace                              |
| Monthly_Income                          | Monthly salary                                               |
| Education_Level                         | Education qualification level                                |
| Age                                     | Age of employee                                              |
| Num_Companies_Worked                    | Number of companies worked previously                        |
| Employee_Role                           | Encoded job role                                             |
| Annual_Bonus                            | Bonus received annually                                      |
| Training_Hours                          | Training hours attended                                      |
| Department                              | Encoded department                                           |
| Annual_Bonus_Squared                    | Engineered feature (bonus <sup>2</sup> )                     |
| Annual_Bonus_Training_Hours_Interaction | Interaction feature between annual bonus and training hours. |
| <b>Employee_Turnover (Target)</b>       | 1 = Employee Left, 0 = Stayed                                |

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